5.5.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.

I need help. I got it. I could help someone.

Identifying and defining the variable for a real-world context	
Writing a multi-step equation to represent a context	
Solving multi-step one-variable linear equations	
Interpreting the meaning of the solution in a given context	

Unit 5, Lesson 6: Exploring Solutions to Systems of Equations

Benchmark

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

Learning Target

☐ I can relate the solution of a system of equations to its graph and equations.

5.6.1 Warm-Up: Mindsets in Math

1. Some people believe the myth that math intelligence is something people are only born with. This is called having a "fixed mindset." Others recognize that math intelligence is something anyone can develop, which is called having a "growth mindset."

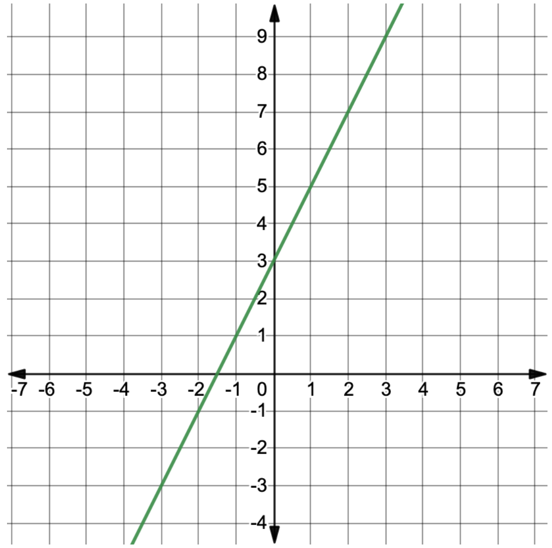
   a. Describe something you have learned in math this year that you didn't initially understand, but you do now.

   b. What helped you learn it?

   c. How does your mindset regarding math intelligence impact the way you learn and face challenges in math class?

5.6.2 Exploration: Solutions to Systems of Equations

1. The graph of the equation $y = 2x + 3$ is shown.



   a. Mark the point $(2, 7)$ on the graph. What do you notice?

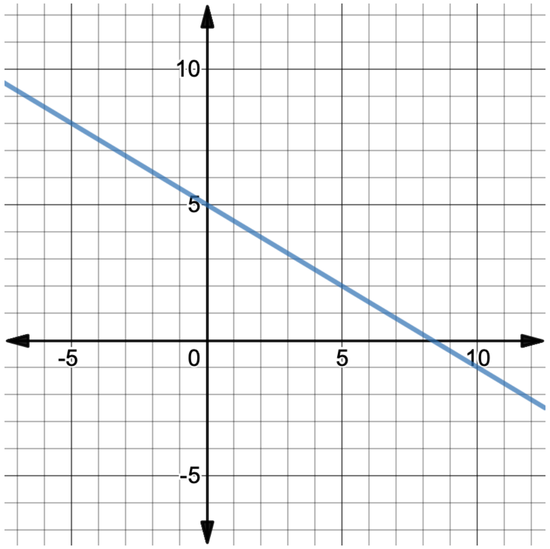
   b. Complete the statement.

   Since the point $(2, 7)$ lies on the line of $y = 2x + 3$, it is considered a(n)
   - ○ intercept
   - ○ solution
   - ○ vertex

   to the equation.

   c. Verify algebraically that point $(2, 7)$ is a solution to the equation $y = 2x + 3$.

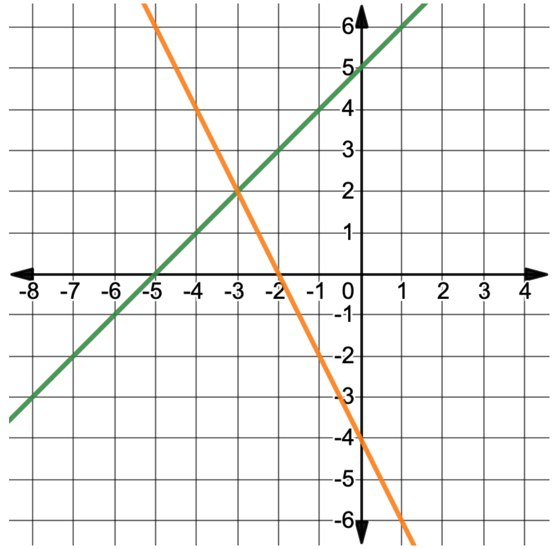
2. The graph of the equation $y = -0.6x + 5$ is shown.



Work with your partner to find two different solutions to the equation $y = -0.6x + 5$. Then, verify algebraically that each is a solution.

Solution 1: (____ , ____)	**Solution 2: (____ , ____)**
Algebraic Verification	Algebraic Verification

3. The graphs of two linear equations are shown.



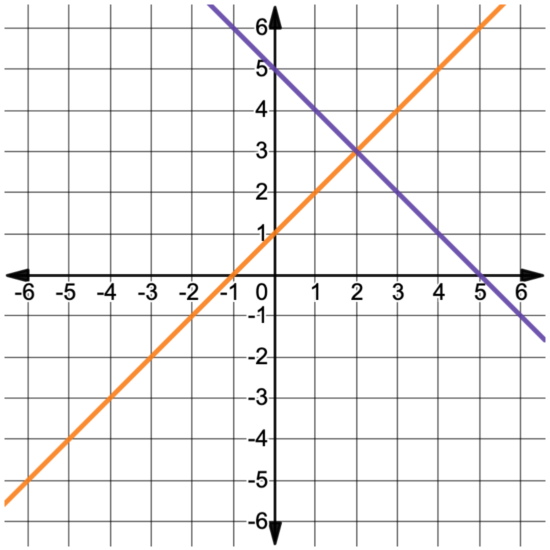
a. Discuss with your partner whether or not there is a common solution for both linear equations shown on the graph.

b. Complete the statement.

The point (______, ______) lies on both lines, so it is considered a solution of both linear equations.

A ***system of equations*** is a set of two or more equations. Each equation contains two or more variables. The solution is all the values for the variables that make all equations in the system true.

4. The graph of the system of equations, $\begin{cases} y = x + 1 \\ y = -x + 5 \end{cases}$, is shown.

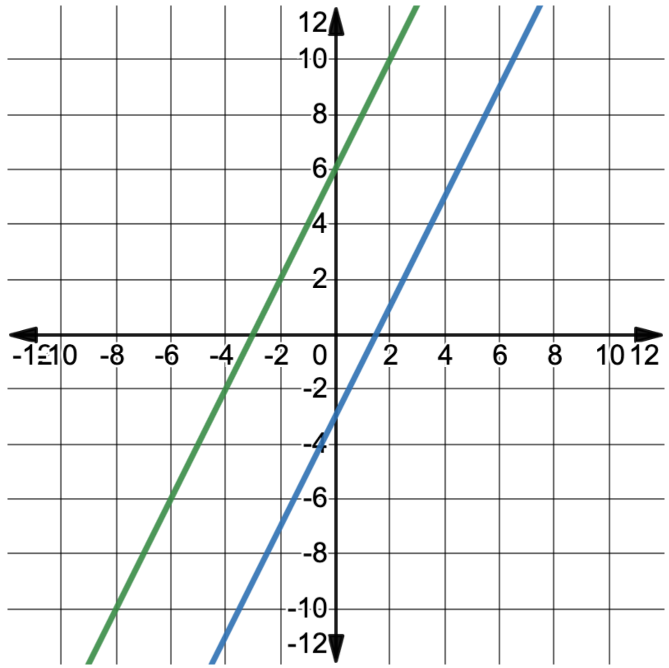


a. Work with your partner to determine the solution to the system of equations from the graph.

b. Verify algebraically that your solution satisfies both equations in the system.

$y = x + 1$	$y = -x + 5$

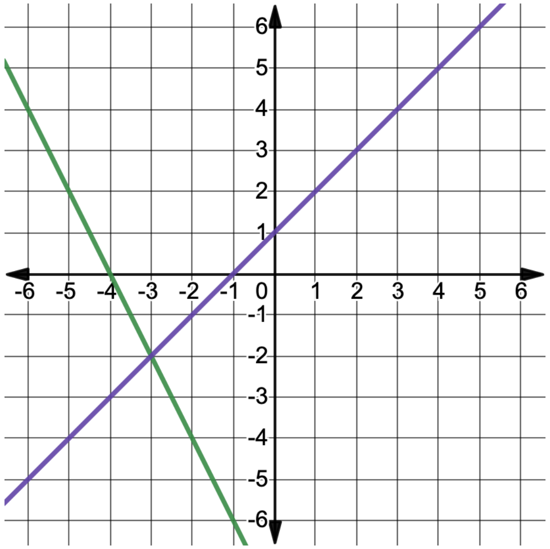
5. The graph of a system of equations is shown.



Work with your partner to describe the solution of the system.

5.6.3 Exploration: Equivalent Systems of Equations

1. Chloe, Armin, and Olivia are all solving the system of equations, $\begin{cases} y = -2x - 8 \\ y = x + 1 \end{cases}$. The graph of the system is shown.



a. What is the solution to this system?

Each student rewrote the original system of equations in different ways. Their new systems and how they generated them are shown in the table.

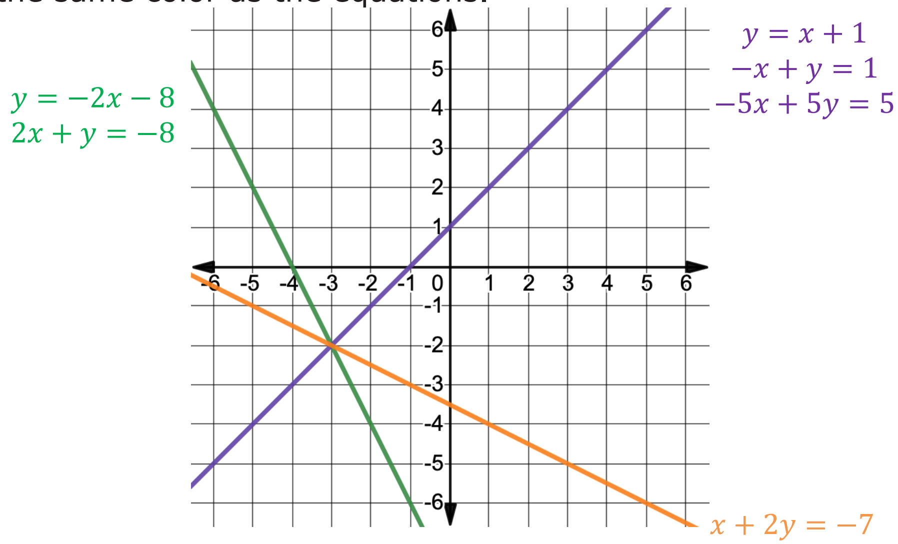
Student	Student's Work	New System
Chloe	$y = -2x - 8$ $\underline{+2x \quad +2x}$ $2x + y = -8$ $y = x + 1$ $\underline{-x \quad -x}$ $-x + y = 1$	$\begin{cases} 2x + y = -8 \\ -x + y = 1 \end{cases}$
Armin	$y = -2x - 8$ $\underline{+(y = x + 1)}$ $2y = -x - 7$ $\underline{+x \quad +x}$ $x + 2y = -7$ $y = x + 1$ $\underline{-x \quad -x}$ $-x + y = 1$	$\begin{cases} x + 2y = -7 \\ -x + y = 1 \end{cases}$
Olivia	$y = -2x - 8$ $\underline{+2x \quad +2x}$ $2x + y = -8$ $5(y = x + 1)$ $5y = 5x + 5$ $\underline{-5x \quad -5x}$ $-5x + 5y = 5$	$\begin{cases} 2x + y = -8 \\ -5x + 5y = 5 \end{cases}$

b. Chloe rearranged her equations. Explain what Armin did to generate his new system.

c. Explain how Olivia generated her new system.

d. With your partner, discuss whose new system you think will result in the same solution as the original system. Then, summarize your discussion.

The graphs of the original system of equations and the newly created systems are shown on the same coordinate grid. The graphs that represent each equation are shown in the same color as the equations.



The graphs of $y = x + 1$, $-x + y = 1$, and $-5x + 5y = 5$ are on the same line, shown in purple on the coordinate grid. Therefore, these equations are **equivalent**, not equal. To be equal, the equations would be exactly the same.

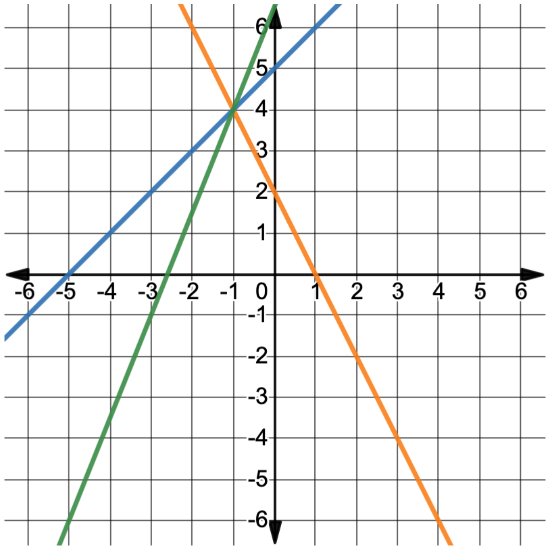
When two entities have the same result, they are **equivalent**. Systems of equations with the same solution are **equivalent systems**.

e. Complete the statement.

Chloe, Armin, and Olivia all generated systems of equations ○ equal ○ equivalent to the original system because all of the systems result in ○ the same ○ different solution(s).

5.6.4 Bring It Together: Systems of Equations

1. The graphs of the systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are shown on the same coordinate plane.



Complete the statements.

a. The solution of any system of equations is the

- ○ point(s) of intersection.
- ○ slope.
- ○ y-intercept.

b. The solution of the system $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ is (_____, _____).

c. The solution of the system $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ is (_____, _____).

d. The systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are __________________ systems because they have the ____________ solution.

2. Verify algebraically that the solution satisfies all of the equations included in either system.

$2x + y = 2$	$x - y = -5$	$5x - 2y = -13$

5.6.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Explorations that I knew were …	Something I wondered during the Explorations was …	Something I learned during the Explorations was …

Unit 5, Lesson 7: Solving Systems of Equations by Graphing

Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

- ☐ I can solve systems of equations by graphing, while noting constraints for a real-world context.
- ☐ I can interpret the solution to a system of equations in a real-world context.
- ☐ I can interpret the coefficients of a given system representing a real-world context.